CHAPTER C

Assessments and annual preparation

The T1 and T2 versions are available to students. The A and B versions and all the solutions are reserved for the teacher. The papers are original; they are not presented as official past papers. The approaches change between versions: direct or inverse model, optimisation or relative position, direct or reversed tree, projection or locus.

EV1-T1 – Common test

Practice paper. Chapters 1 to 4.

Duration: 2 h. No calculator. Marked out of 20. 14 November 2026.

Part 1 – Fluency (6 points)

Twelve questions, 0.5 point each. Only one correct answer; no justification required. An incorrect or missing answer scores 0, with no loss of points.

1. Calculate $3 + 2 \times 3$.
   A. 12 B. 9 C. 8 D. 15
2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{3}$ B. $\frac{2}{5}$ C. $\frac{5}{6}$ D. $\frac{1}{6}$
3. Calculate $\dfrac{2^3}{2^1}$.
   A. 4 B. 8 C. 2 D. 1
4. A value of 50 increases by 20%. Final value?
   A. 10 B. 40 C. 60 D. 70
5. A value of 80 is obtained after a 20% decrease. Initial value?
   A. 60 B. 64 C. 96 D. 100
6. A 25% increase then a 20% decrease: overall coefficient?
   A. $\frac{4}{5}$ B. $\frac{21}{20}$ C. $\frac{9}{10}$ D. 1
7. The solution of $2x + 3 = 0$ is
   A. $\frac{3}{2}$ B. $\frac{-3}{2}$ C. 6 D. -6
8. The slope between $(0, 3)$ and $(2, 9)$ is
   A. 2 B. -3 C. 6 D. 3
9. The mean of $3, 5, 7$ is
   A. 6 B. 15 C. 3 D. 5
10. A sector of $90°$ represents what fraction of the total?
    A. $\frac{1}{2}$ B. $\frac{1}{4}$ C. $\frac{3}{4}$ D. $\frac{1}{3}$
11. A probability is $\dfrac{2}{5}$. Its complement is
    A. $\frac{2}{5}$ B. $\frac{3}{5}$ C. $\frac{5}{2}$ D. $\frac{1}{5}$
12. For $f(x) = x^2 + 3$, $f(-2)$ is
    A. 7 B. 5 C. 12 D. -1

Part 2 – Independent exercises (14 points)

Every step must be justified. Exact results are expected unless stated otherwise.

Exercise 1 – A book collection (5 points)

A library has 120 books at the end of the reference month, called month 0. At the end of each following month, it adds 18 books to its stock; no book is removed. Let u_n be the number of books available at the end of month n, for every natural number n.

1. Determine the type of (u_n), its common difference and its expression in terms of n. (1 pt)
2. Calculate u_8, then $u_0 + u_1 + \cdots + u_8$. Explain what this sum accumulates and why it does not count books that are all distinct. (2 pts)
3. Determine the first month in which the stock strictly exceeds 300 books. Check the previous month. (2 pts)

Exercise 2 – Area and choice of form (5 points)

A rectangle has a perimeter of 28 m. If one of its sides measures x metres, the other measures $14 - x$ metres. Its area is modelled by the function A defined on $[0, 14]$ by $A(x) = x(14 - x)$, in square metres. The endpoints represent degenerate rectangles.

1. Expand, then write A in vertex form. (1 pt)
2. Determine the maximum area and the corresponding value of x without a derivative. (2 pts)
3. Solve $A(x) \geq 40$ on the domain. (2 pts)

Exercise 3 – A right angle (4 points)

The plane is equipped with an orthonormal frame. Given the points $A(0, 1)$, $B(3, 2)$ and $C(-1, 4)$. Lengths are expressed in the unit of the frame and areas in area units.

1. Calculate the coordinates of the vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$. (1 pt)
2. Prove that these vectors are orthogonal and state the nature of the triangle. (1 pt)
3. Calculate the length BC then the area of the triangle ABC. (2 pts)

EV1-T2 – Common test

Practice paper. Chapters 1 to 4.

Duration: 2 h. No calculator. Marked out of 20. 14 November 2026.

Part 1 – Fluency (6 points)

Twelve questions, 0.5 point each. Only one correct answer; no justification required. An incorrect or missing answer scores 0, with no loss of points.

1. Calculate $4 + 2 \times 3$.
   A. 18 B. 9 C. 13 D. 10
2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{3}$ B. $\frac{2}{5}$ C. $\frac{5}{6}$ D. $\frac{1}{6}$
3. Calculate $\dfrac{2^4}{2^2}$.
   A. 2 B. 4 C. 1 D. 9
4. A value of 50 increases by 20%. Final value?
   A. 70 B. 40 C. 10 D. 60
5. A value of 80 is obtained after a 20% decrease. Initial value?
   A. 100 B. 96 C. 64 D. 60
6. A 25% increase then a 20% decrease: overall coefficient?
   A. 1 B. $\frac{21}{20}$ C. $\frac{4}{5}$ D. $\frac{9}{10}$
7. The solution of $2x + 4 = 0$ is
   A. 2 B. 8 C. -2 D. -8
8. The slope between $(0, 4)$ and $(2, 10)$ is
   A. 3 B. -3 C. 6 D. 2
9. The mean of $4, 6, 8$ is
   A. 18 B. 6 C. 7 D. 4
10. A sector of $90°$ represents what fraction of the total?
    A. $\frac{1}{3}$ B. $\frac{3}{4}$ C. $\frac{1}{4}$ D. $\frac{1}{2}$
11. A probability is $\dfrac{2}{5}$. Its complement is
    A. $\frac{5}{2}$ B. $\frac{2}{5}$ C. $\frac{1}{5}$ D. $\frac{3}{5}$
12. For $f(x) = x^2 + 4$, $f(-2)$ is
    A. 6 B. 16 C. 8 D. 0

Part 2 – Independent exercises (14 points)

Every step must be justified. Exact results are expected unless stated otherwise.

Exercise 1 – Two growth processes (5 points)

We compare two models for the output of a workshop. On day 0, each forecasts 30 items. Model A forecasts 10 extra items each day; model B forecasts a daily doubling. The outputs on day n are therefore $a_n = 30 + 10n$ and $b_n = 30 \times 2^n$, for $n \in \mathbb{N}$. A running total here means the sum of the outputs of the days indicated.

1. State the type of each sequence and calculate the outputs on day 2. (1 pt)
2. Calculate the cumulative outputs for days 0 to 3 in each model. (2 pts)
3. Check that model B exceeds model A on day 1, then explain why this gap stays positive on all following days by comparing the daily increments. (2 pts)

Exercise 2 – Imposed roots (5 points)

We look for a polynomial function f of degree 2, defined on $\mathbb{R}$. Its curve crosses the x-axis at the x-coordinates -1 and 5, and it passes through the point $(0, 10)$. This information must be enough to determine the function.

1. Give its factorised form. (1 pt)
2. Determine its sign then solve $f(x) \leq 0$. (2 pts)
3. Give its vertex form and its maximum. (2 pts)

Exercise 3 – A controlled approximation (4 points)

The function f is defined on $\mathbb{R}$ by $f(x) = x^2 + 1$. We wish to approximate $f(3.02)$ using the tangent to its curve at the point with x-coordinate 3. The calculations are done without a calculator.

1. For $h \neq 0$, calculate and simplify the rate of change between 3 and $3 + h$. (1 pt)
2. Deduce $f'(3)$, then the equation of the tangent at the point with x-coordinate 3. (1 pt)
3. Use the tangent to approximate $f(3.02)$. Then calculate the error "exact value minus approximation". (2 pts)

EV2-T1 – Mock exam 1

Practice paper. Chapters 1 to 7.

Duration: 2 h. No calculator. Marked out of 20. 14–16 January 2027.

Part 1 – Fluency (6 points)

Twelve questions, 0.5 point each. Only one correct answer; no justification required. An incorrect or missing answer scores 0, with no loss of points.

1. Calculate $2 + 2 \times 3$.
   A. 8 B. 12 C. 11 D. 7
2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{1}{6}$ B. $\frac{5}{3}$ C. $\frac{2}{5}$ D. $\frac{5}{6}$
3. Calculate $\dfrac{2^2}{2^0}$.
   A. 2 B. 7 C. 1 D. 4
4. A value of 50 increases by 20%. Final value?
   A. 40 B. 60 C. 70 D. 10
5. A value of 80 is obtained after a 20% decrease. Initial value?
   A. 96 B. 100 C. 64 D. 60
6. A 25% increase then a 20% decrease: overall coefficient?
   A. $\frac{4}{5}$ B. $\frac{9}{10}$ C. 1 D. $\frac{21}{20}$
7. The solution of $2x + 2 = 0$ is
   A. -4 B. -1 C. 1 D. 4
8. The slope between $(0, 2)$ and $(2, 8)$ is
   A. 6 B. 2 C. 3 D. -3
9. The mean of $2, 4, 6$ is
   A. 4 B. 2 C. 12 D. 5
10. A sector of $90°$ represents what fraction of the total?
    A. $\frac{1}{3}$ B. $\frac{3}{4}$ C. $\frac{1}{4}$ D. $\frac{1}{2}$
11. A probability is $\dfrac{2}{5}$. Its complement is
    A. $\frac{2}{5}$ B. $\frac{5}{2}$ C. $\frac{1}{5}$ D. $\frac{3}{5}$
12. For $f(x) = x^2 + 2$, $f(-2)$ is
    A. 4 B. 8 C. 6 D. -2

Part 2 – Independent exercises (14 points)

Every step must be justified. Exact results are expected unless stated otherwise.

Exercise 1 – Manufacturing and quality control (5 points)

Two workshops A and B produce 30 percent and 70 percent respectively of a company's items. Among the items from A, 10 percent are defective; this proportion is 20 percent for B. An item is chosen at random from the total production. The events A and D mean respectively "comes from workshop A" and "is defective".

1. Construct a weighted tree whose first level distinguishes the two workshops. (1 pt)
2. Calculate $P(D)$, then the probability that a defective item comes from A. (2 pts)
3. Are the events A and D independent? Justify numerically. (2 pts)

Exercise 2 – Minimum of a quotient (5 points)

The function f is defined on the interval $(0, +\infty)$ by $f(x) = x + \dfrac{9}{x}$. We study its variation and certain tangents to its curve. The coordinates requested are to be given in exact form.

1. Calculate and factorise the derivative. (1 pt)
2. Study the variation and give the minimum. (2 pts)
3. Determine the points where the tangent has slope $\dfrac{3}{4}$. (2 pts)

Exercise 3 – Projection and circle (4 points)

In an orthonormal frame, consider the point $P(5, 2)$ and the line d with equation $x + y - 3 = 0$. Let H be the orthogonal projection of P onto d.

1. Give a nonzero normal vector to d. (1 pt)
2. Determine the coordinates of H and check the conditions of membership and orthogonality. (1 pt)
3. Give an equation of the circle with centre P passing through H. (2 pts)

EV2-T2 – Mock exam 1

Practice paper. Chapters 1 to 7.

Duration: 2 h. No calculator. Marked out of 20. 14–16 January 2027.

Part 1 – Fluency (6 points)

Twelve questions, 0.5 point each. Only one correct answer; no justification required. An incorrect or missing answer scores 0, with no loss of points.

1. Calculate $3 + 2 \times 3$.
   A. 12 B. 8 C. 15 D. 9
2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{2}{5}$ B. $\frac{1}{6}$ C. $\frac{5}{3}$ D. $\frac{5}{6}$
3. Calculate $\dfrac{2^3}{2^1}$.
   A. 4 B. 2 C. 8 D. 1
4. A value of 50 increases by 20%. Final value?
   A. 40 B. 70 C. 60 D. 10
5. A value of 80 is obtained after a 20% decrease. Initial value?
   A. 64 B. 96 C. 100 D. 60
6. A 25% increase then a 20% decrease: overall coefficient?
   A. $\frac{9}{10}$ B. $\frac{21}{20}$ C. $\frac{4}{5}$ D. 1
7. The solution of $2x + 3 = 0$ is
   A. 6 B. -6 C. $\frac{3}{2}$ D. $\frac{-3}{2}$
8. The slope between $(0, 3)$ and $(2, 9)$ is
   A. -3 B. 2 C. 6 D. 3
9. The mean of $3, 5, 7$ is
   A. 3 B. 5 C. 6 D. 15
10. A sector of $90°$ represents what fraction of the total?
    A. $\frac{1}{4}$ B. $\frac{1}{3}$ C. $\frac{3}{4}$ D. $\frac{1}{2}$
11. A probability is $\dfrac{2}{5}$. Its complement is
    A. $\frac{2}{5}$ B. $\frac{5}{2}$ C. $\frac{1}{5}$ D. $\frac{3}{5}$
12. For $f(x) = x^2 + 3$, $f(-2)$ is
    A. 12 B. 7 C. -1 D. 5

Part 2 – Independent exercises (14 points)

Every step must be justified. Exact results are expected unless stated otherwise.

Exercise 1 – Two draws (5 points)

An urn contains four red balls and two blue balls, indistinguishable by touch. Two balls are drawn in succession at random, without replacing the first. At each draw, the balls still present are equally likely. The third question proposes another protocol, with replacement.

1. Calculate the probability of drawing two red balls. (1 pt)
2. Calculate the probability of drawing one ball of each colour, taking into account the two possible orders. (2 pts)
3. The first ball is now replaced before the second draw. Calculate the probability of drawing one ball of each colour and compare it with that of the protocol without replacement. (2 pts)

Exercise 2 – A box (5 points)

Squares of side x cm are cut from the four corners of a square sheet of side 10 cm. Folding up the edges forms a box without a lid. Its volume, in cubic centimetres, is $V(x) = x(10 - 2x)^2$. We study this model on $[0, 5]$, endpoints included even though they are degenerate.

1. Calculate V' in factorised form. (1 pt)
2. Study the variation. (2 pts)
3. Determine the exact maximum volume and justify that the endpoints do not give the maximum. (2 pts)

Exercise 3 – Recognition and intersection (4 points)

In an orthonormal frame, consider the set $\mathcal{C}$ of points $M(x, y)$ satisfying $x^2 + y^2 + 4x - 2y - 4 = 0$. After identifying it, we will study its intersections with a line and a tangent.

1. Show that $\mathcal{C}$ is a circle and give its centre and radius. (1 pt)
2. Determine all the points common to $\mathcal{C}$ and the line $x = -2$. (1 pt)
3. Write the equation of the tangent to the circle at the intersection point with the greater y-coordinate. Justify its direction. (2 pts)

EV3-T1 – Mock exam 2

Practice paper. Programme studied in chapters 1 to 10.

Duration: 2 h. No calculator. Marked out of 20. 1–3 April 2027.

Part 1 – Fluency (6 points)

Twelve questions, 0.5 point each. Only one correct answer; no justification required. An incorrect or missing answer scores 0, with no loss of points.

1. Calculate $6 + 2 \times 3$.
   A. 12 B. 24 C. 11 D. 15
2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{2}{5}$ B. $\frac{5}{3}$ C. $\frac{5}{6}$ D. $\frac{1}{6}$
3. Calculate $\dfrac{2^6}{2^4}$.
   A. 1 B. 11 C. 2 D. 4
4. A value of 50 increases by 20%. Final value?
   A. 10 B. 40 C. 70 D. 60
5. A value of 80 is obtained after a 20% decrease. Initial value?
   A. 64 B. 96 C. 100 D. 60
6. A 25% increase then a 20% decrease: overall coefficient?
   A. $\frac{21}{20}$ B. $\frac{9}{10}$ C. 1 D. $\frac{4}{5}$
7. The solution of $2x + 6 = 0$ is
   A. 12 B. -12 C. -3 D. 3
8. The slope between $(0, 6)$ and $(2, 12)$ is
   A. -3 B. 3 C. 2 D. 6
9. The mean of $6, 8, 10$ is
   A. 9 B. 8 C. 24 D. 6
10. A sector of $90°$ represents what fraction of the total?
    A. $\frac{1}{2}$ B. $\frac{1}{3}$ C. $\frac{1}{4}$ D. $\frac{3}{4}$
11. A probability is $\dfrac{2}{5}$. Its complement is
    A. $\frac{5}{2}$ B. $\frac{2}{5}$ C. $\frac{3}{5}$ D. $\frac{1}{5}$
12. For $f(x) = x^2 + 6$, $f(-2)$ is
    A. 10 B. 8 C. 24 D. 2

Part 2 – Independent exercises (14 points)

Every step must be justified. Exact results are expected unless stated otherwise.

Exercise 1 – A compound gain (5 points)

A game consists of two independent trials, each with success probability $\frac{1}{2}$. Each success earns 3 monetary units; taking part costs 2 units. Let X be the net gain, that is, the total received minus the entry fee paid.

1. Establish the probability distribution of the net gain X. (1 pt)
2. Calculate its expectation and variance. (2 pts)
3. Determine the entry fee that would make the game fair. Explain what the standard deviation of the net gain would become. (2 pts)

Exercise 2 – Peak of a model (5 points)

A quantity is modelled for $t \geq 0$ by the function $f(t) = (t+2)e^{-t}$. The time t is expressed in a fixed unit. We look for its maximum value, then an approximation near the initial time.

1. Differentiate f. (1 pt)
2. Study the variation and give the maximum. (2 pts)
3. Write the tangent at 0 and approximate $f(0.1)$. (2 pts)

Exercise 3 – The unit circle (4 points)

The plane is equipped with an orthonormal frame. On the unit circle with centre O, the points A and B are the images of the real numbers $\frac{\pi}{6}$ and $\frac{5\pi}{6}$ respectively.

1. Give their coordinates. (1 pt)
2. Calculate AB. (1 pt)
3. Give the equation of their perpendicular bisector and justify geometrically. (2 pts)

EV3-T2 – Mock exam 2

Practice paper. Programme studied in chapters 1 to 10.

Duration: 2 h. No calculator. Marked out of 20. 1–3 April 2027.

Part 1 – Fluency (6 points)

Twelve questions, 0.5 point each. Only one correct answer; no justification required. An incorrect or missing answer scores 0, with no loss of points.

1. Calculate $2 + 2 \times 3$.
   A. 11 B. 7 C. 12 D. 8
2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{1}{6}$ B. $\frac{2}{5}$ C. $\frac{5}{3}$ D. $\frac{5}{6}$
3. Calculate $\dfrac{2^2}{2^0}$.
   A. 7 B. 4 C. 2 D. 1
4. A value of 50 increases by 20%. Final value?
   A. 40 B. 70 C. 10 D. 60
5. A value of 80 is obtained after a 20% decrease. Initial value?
   A. 96 B. 60 C. 100 D. 64
6. A 25% increase then a 20% decrease: overall coefficient?
   A. $\frac{4}{5}$ B. $\frac{21}{20}$ C. 1 D. $\frac{9}{10}$
7. The solution of $2x + 2 = 0$ is
   A. -1 B. 4 C. -4 D. 1
8. The slope between $(0, 2)$ and $(2, 8)$ is
   A. 3 B. 2 C. 6 D. -3
9. The mean of $2, 4, 6$ is
   A. 2 B. 4 C. 5 D. 12
10. A sector of $90°$ represents what fraction of the total?
    A. $\frac{1}{4}$ B. $\frac{1}{2}$ C. $\frac{1}{3}$ D. $\frac{3}{4}$
11. A probability is $\dfrac{2}{5}$. Its complement is
    A. $\frac{5}{2}$ B. $\frac{3}{5}$ C. $\frac{1}{5}$ D. $\frac{2}{5}$
12. For $f(x) = x^2 + 2$, $f(-2)$ is
    A. 6 B. -2 C. 4 D. 8

Part 2 – Independent exercises (14 points)

Every step must be justified. Exact results are expected unless stated otherwise.

Exercise 1 – Simulation and mean (5 points)

The random variable X takes the values 0, 2 and 6 with respective probabilities $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{4}$. We then simulate samples of 25 independent realisations of this variable and let m be the mean observed in a sample.

1. Calculate the expectation $E(X)$. (1 pt)
2. Calculate $V(X)$ then the exact standard deviation $\sigma(X)$. (2 pts)
3. Write the interval of values of m satisfying $|m - E(X)| \leq \dfrac{2\sigma(X)}{5}$. Explain why a simulation on a finite number of samples does not guarantee an exact coverage percentage. (2 pts)

Exercise 2 – Comparing two models (5 points)

Define on $\mathbb{R}$ the functions $f(x) = e^{2x}$ and $g(x) = 1 + 2x$, then their difference $h = f - g$. The last question considers the values of f at integer times, starting from index 0.

1. Calculate the derivative of $h = f - g$. (1 pt)
2. Study h and compare the curves. (2 pts)
3. For $u_n = e^{2n}$, give the common ratio and the sum of the first four terms. (2 pts)

Exercise 3 – A displaced circle (4 points)

In an orthonormal frame, consider the circle $\mathcal{C}$ with equation $(x-1)^2 + (y+1)^2 = 4$ and the line d with equation $y = -1$. Call A and B their points of intersection, A being the one with the smaller x-coordinate.

1. Give the centre and radius. (1 pt)
2. Determine the intersections with the line. (1 pt)
3. For M on the circle, calculate $\overrightarrow{MA} \cdot \overrightarrow{MB}$ where A and B are these intersections; justify. (2 pts)

EV4-T1 – EAM preparation

Practice paper. Whole programme.

Duration: 2 h. No calculator. Marked out of 20. Practice sessions in May and June; EAM date for the zone to be confirmed.

Part 1 – Fluency (6 points)

Twelve questions, 0.5 point each. Only one correct answer; no justification required. An incorrect or missing answer scores 0, with no loss of points.

1. Calculate $5 + 2 \times 3$.
   A. 14 B. 11 C. 21 D. 10
2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{3}$ B. $\frac{5}{6}$ C. $\frac{1}{6}$ D. $\frac{2}{5}$
3. Calculate $\dfrac{2^5}{2^3}$.
   A. 1 B. 4 C. 10 D. 2
4. A value of 50 increases by 20%. Final value?
   A. 40 B. 10 C. 60 D. 70
5. A value of 80 is obtained after a 20% decrease. Initial value?
   A. 60 B. 96 C. 100 D. 64
6. A 25% increase then a 20% decrease: overall coefficient?
   A. $\frac{4}{5}$ B. 1 C. $\frac{9}{10}$ D. $\frac{21}{20}$
7. The solution of $2x + 5 = 0$ is
   A. $\frac{5}{2}$ B. 10 C. $\frac{-5}{2}$ D. -10
8. The slope between $(0, 5)$ and $(2, 11)$ is
   A. 3 B. -3 C. 6 D. 2
9. The mean of $5, 7, 9$ is
   A. 7 B. 21 C. 5 D. 8
10. A sector of 90° represents what fraction of the total?
    A. $\frac{1}{4}$ B. $\frac{3}{4}$ C. $\frac{1}{3}$ D. $\frac{1}{2}$
11. A probability is $\dfrac{2}{5}$. Its complement is
    A. $\frac{3}{5}$ B. $\frac{1}{5}$ C. $\frac{2}{5}$ D. $\frac{5}{2}$
12. For $f(x) = x^2 + 5$, $f(-2)$ is
    A. 20 B. 7 C. 1 D. 9

Part 2 – Independent exercises (14 points)

Every step must be justified. Exact results are expected unless stated otherwise.

Exercise 1 – Choosing a distribution (5 points)

We compare two games. The first gives a gain $X = 2$ with certainty. The second gives a gain $Y = -1$ with probability $\frac{2}{3}$ and a gain $Y = 8$ with probability $\frac{1}{3}$. The gains are expressed in the same monetary unit; a negative value represents a loss.

1. Calculate $E(X)$ and $E(Y)$. (1 pt)
2. Calculate $V(Y)$ and compare the dispersions of the two gains. (2 pts)
3. Both games are played and the gains added. Calculate $E(X+Y)$, citing the property used. (2 pts)

Exercise 2 – Discrete threshold and continuous model (5 points)

A quantity is modelled by $Q(t) = 64e^{-t}$ for $t \geq 0$. It is observed at integer times $n \in \mathbb{N}$ and we set $u_n = Q(n)$. For comparisons without a calculator, the bounds $e^2 < 8 < e^3$ are given.

1. Study the direction of variation. (1 pt)
2. Determine the first integer n such that $Q(n) < 8$ using the bounds provided. (2 pts)
3. Give the type of $u_n = Q(n)$ and the expression of $u_0 + \cdots + u_3$. (2 pts)

Exercise 3 – Equation and circle (4 points)

In an orthonormal frame, consider the set $\mathcal{C}$ defined by $x^2 + y^2 - 2x - 4y + 1 = 0$. The task is to determine its characteristics, then to identify a particular tangent.

1. Show that $\mathcal{C}$ is a circle and state its centre and radius. (1 pt)
2. Determine its points of intersection with the line with equation $x = 1$. (1 pt)
3. Determine the equation of the tangent to the circle at the point with the greatest y-coordinate and justify. (2 pts)

EV4-T2 – EAM preparation

Practice paper. Whole programme.

Duration: 2 h. No calculator. Marked out of 20. Practice sessions in May and June; EAM date for the zone to be confirmed.

Part 1 – Fluency (6 points)

Twelve questions, 0.5 point each. Only one correct answer; no justification required. An incorrect or missing answer scores 0, with no loss of points.

1. Calculate $6 + 2 \times 3$.
   A. 24 B. 15 C. 11 D. 12
2. Calculate $\frac{1}{2} + \frac{1}{3}$.
   A. $\frac{5}{6}$ B. $\frac{2}{5}$ C. $\frac{5}{3}$ D. $\frac{1}{6}$
3. Calculate $\dfrac{2^6}{2^4}$.
   A. 2 B. 4 C. 1 D. 11
4. A value of 50 increases by 20%. Final value?
   A. 60 B. 10 C. 70 D. 40
5. A value of 80 is obtained after a 20% decrease. Initial value?
   A. 64 B. 100 C. 96 D. 60
6. A 25% increase then a 20% decrease: overall coefficient?
   A. $\frac{4}{5}$ B. $\frac{21}{20}$ C. 1 D. $\frac{9}{10}$
7. The solution of $2x + 6 = 0$ is
   A. -3 B. 12 C. 3 D. -12
8. The slope between $(0, 6)$ and $(2, 12)$ is
   A. 2 B. -3 C. 3 D. 6
9. The mean of $6, 8, 10$ is
   A. 6 B. 24 C. 8 D. 9
10. A sector of 90° represents what fraction of the total?
    A. $\frac{1}{3}$ B. $\frac{1}{2}$ C. $\frac{3}{4}$ D. $\frac{1}{4}$
11. A probability is $\dfrac{2}{5}$. Its complement is
    A. $\frac{5}{2}$ B. $\frac{2}{5}$ C. $\frac{1}{5}$ D. $\frac{3}{5}$
12. For $f(x) = x^2 + 6$, $f(-2)$ is
    A. 10 B. 8 C. 24 D. 2

Part 2 – Independent exercises (14 points)

Every step must be justified. Exact results are expected unless stated otherwise.

Exercise 1 – A reversed tree (5 points)

Two events A and B satisfy $P(A) = \dfrac{3}{5}$, $P_A(B) = \dfrac{1}{3}$ and $P_{\overline{A}}(B) = \dfrac{3}{4}$. We wish to build a tree that starts by distinguishing B and $\overline{B}$, instead of A and $\overline{A}$.

1. Calculate $P(A \cap B)$ and $P(B)$. (1 pt)
2. Calculate $P_B(A)$ and $P_{\overline{B}}(A)$. (2 pts)
3. Construct and complete the weighted tree whose first level distinguishes B and $\overline{B}$, then the second A and $\overline{A}$. (2 pts)

Exercise 2 – Two optimisation methods (5 points)

The function f is defined on the interval $[1, 6]$ by $f(x) = x(8 - x)$. We look for its extrema on this precise domain, then for the values of x for which $f(x)$ reaches at least 15.

1. Give a vertex form and a maximum. (1 pt)
2. Recover the maximum by differentiation. (2 pts)
3. Determine the minimum on the domain and solve $f(x) \geq 15$. (2 pts)

Exercise 3 – Arguments and symmetry (4 points)

In an orthonormal frame with origin O, the points A and B of the unit circle are the images of $\dfrac{7\pi}{6}$ and $\dfrac{11\pi}{6}$. The unoriented angle $\widehat{AOB}$ will be given in radians, between 0 and π.

1. Give their exact coordinates. (1 pt)
2. Calculate $\overrightarrow{OA} \cdot \overrightarrow{OB}$. (1 pt)
3. Determine the unoriented angle AOB then AB. (2 pts)